Introduction to Code-based Cryptography

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Blockstream Research

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github.com/BlockstreamResearch/zk-book



Plan

1 Error-Correcting Codes

2 Reed–Solomon Codes

3 Proximity to a Code

What will we learn today?

- Error-correcting codes: encode k symbols into $n > k$ symbols so that any two codewords differ in many positions.
- Codes are everywhere in cryptography: Shamir's secret sharing **is** a Reed–Solomon code, McEliece encryption, post-quantum candidates.
- In proof systems: a Merkle tree gives only **local access** to a committed vector. Committing to an **encoding** with large distance makes local access sufficient.
- This is the principle behind STARKs, Ligero, Brakedown, Binius: only hash functions, no trusted setup, plausibly post-quantum.

Goal

Linear codes, Reed–Solomon codes $\mathrm{RS}[\mathbb{F}, \Omega, k]$, and the notion of **proximity** to a code — the central tool behind STARKs.

Error-Correcting Codes

Motivation

Fix an **alphabet** Σ (say, $\Sigma = \{0, 1\}$ for binary strings).

Problem

Alice wants to send Bob a message $m \in \Sigma^k$. Alice sits in Germany, while Bob is on the Moon. Whatever message Alice picks, two random letters of m get corrupted. How can Alice transmit the exact message m to Bob?

Idea:

- An **encoder** maps k symbols to $n > k$ symbols, called a **codeword**: we add $n - k$ redundant symbols.
- The receiver gets a **received word** of n symbols, which may differ from the codeword in some positions.
- A **decoder** recovers m using the redundant information.

Transmission over a Noisy Channel

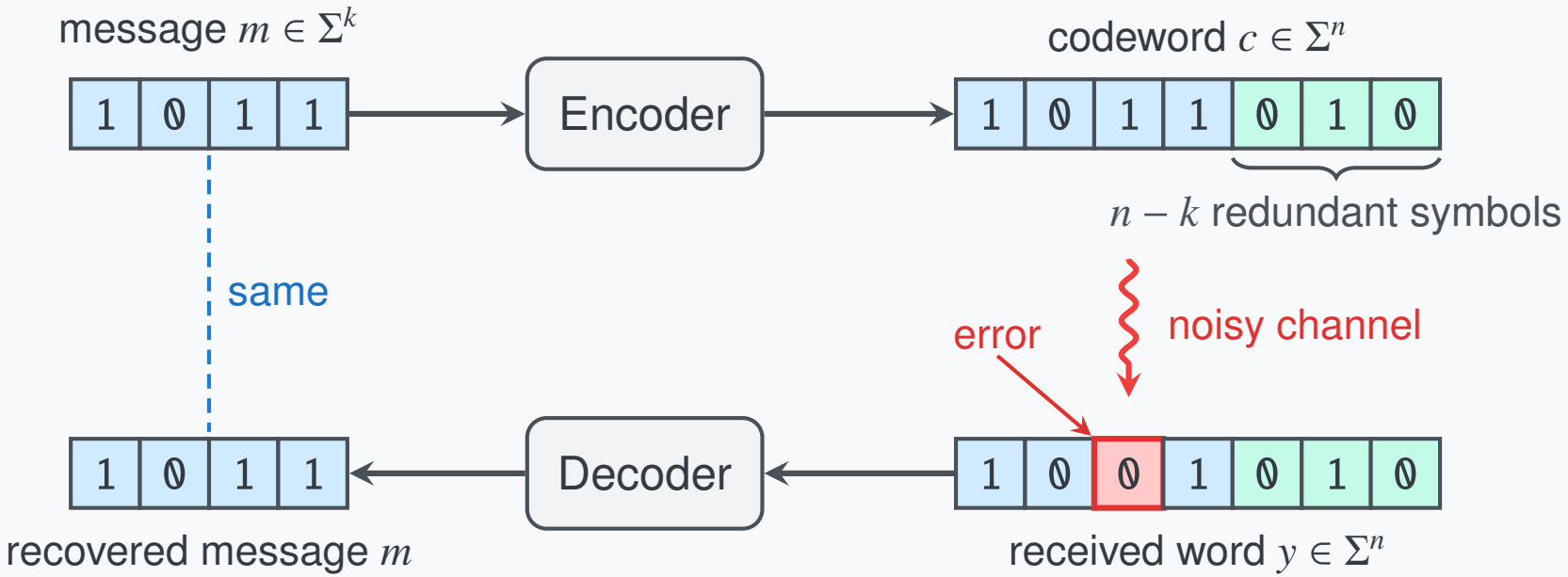


Figure: The encoder expands $k = 4$ symbols into $n = 7$ symbols. The channel corrupts one symbol, and the decoder still recovers the original message.

Code

Definition (Code)

A **code** of block length n over an alphabet Σ is a subset $C \subseteq \Sigma^n$, whose elements are called **codewords**.

- $\#C$ is the number of codewords;
- $k := \log_{\#\Sigma} \#C$ is the number of message symbols.

An **encoding** for C is an injective map $\Sigma^k \to \Sigma^n$ whose image is C.

Question

How do we measure "how much" two words differ?

Hamming Distance

Definition (Hamming distance)

Given $\mathbf{x}, \mathbf{y} \in \Sigma^n$, the **Hamming distance** $\Delta(\mathbf{x}, \mathbf{y})$ is the number of positions in which they differ:

$$\Delta(\mathbf{x}, \mathbf{y}) := \#\{i \in [n] : x_i \neq y_i\}.$$

Example

Let $\Sigma := \{0, 1\}$, $\mathbf{x} := 00000$ and $\mathbf{y} := 10100$. Then $\Delta(\mathbf{x}, \mathbf{y}) = 2$: they differ on positions 0 and 2.

Question #1

Check that (Σ^n, Δ) is a metric space: $\Delta(\mathbf{x}, \mathbf{x}) = 0$, positivity, symmetry, and the triangle inequality $\Delta(\mathbf{x}, \mathbf{z}) \leq \Delta(\mathbf{x}, \mathbf{y}) + \Delta(\mathbf{y}, \mathbf{z})$.

Relative Hamming Distance

Hamming distance alone does not give the full picture: two vectors of length 1000 that differ in 5 places are much more similar than two vectors of length 10 that differ in 5 places.

Definition (Relative Hamming distance)

For $\mathbf{x}, \mathbf{y} \in \Sigma^n$, the **relative Hamming distance** is

$$\delta(\mathbf{x}, \mathbf{y}) := \frac{1}{n}\Delta(\mathbf{x}, \mathbf{y}) \in [0, 1].$$

Question #2

Over $\mathbb{F}_3$, let $\mathbf{x} = (1, 0, 2, 2, 1)$ and $\mathbf{y} = (1, 2, 2, 0, 1)$. What are $\Delta(\mathbf{x}, \mathbf{y})$ and $\delta(\mathbf{x}, \mathbf{y})$?

Minimum Distance

A code's effectiveness depends on the whole set: the worst-case spacing between two codewords is the key metric.

Definition (Minimum distance)

The **minimum distance** of a code $C \subseteq \Sigma^n$ is

$$d_{\min}(C) := \min_{\mathbf{x} \neq \mathbf{y} \in C} \Delta(\mathbf{x}, \mathbf{y}),$$

and the **relative minimum distance** is $\mu(C) := \frac{1}{n} d_{\min}(C)$.

If $d_{\min}(C) = d$, then:

- fewer than d errors never produce another valid codeword;
- with fewer than $d/2$ errors, the corrupted word is strictly closer to the original codeword than to any other.

Unique Decoding

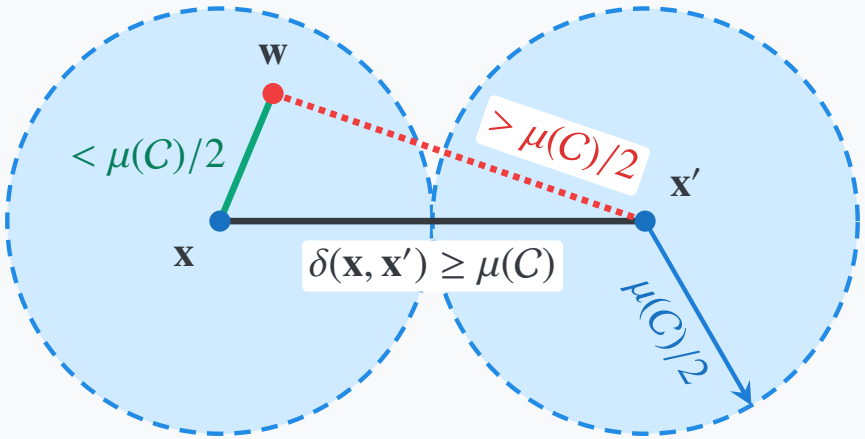
Proposition

Let $C \subseteq \Sigma^n$ be a code. For every $\mathbf{w} \in \Sigma^n$ there exists at most one codeword $\mathbf{x} \in C$ such that $\delta(\mathbf{x}, \mathbf{w}) < \mu(C)/2$.

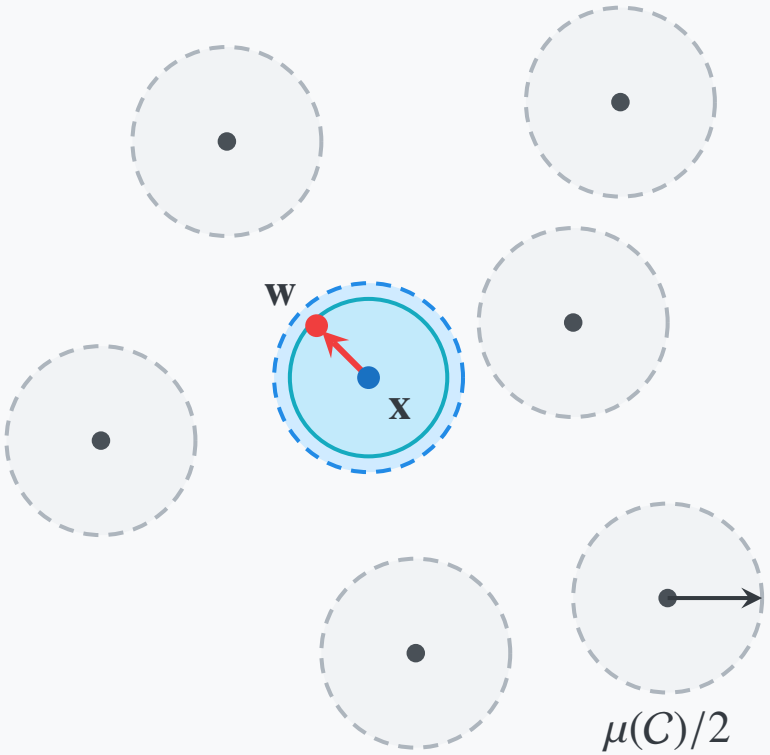
Proof. If $\mathbf{x} \neq \mathbf{x}'$ were both that close, then

$$\delta(\mathbf{x}, \mathbf{x}') \leq \delta(\mathbf{x}, \mathbf{w}) + \delta(\mathbf{w}, \mathbf{x}') < \mu(C),$$

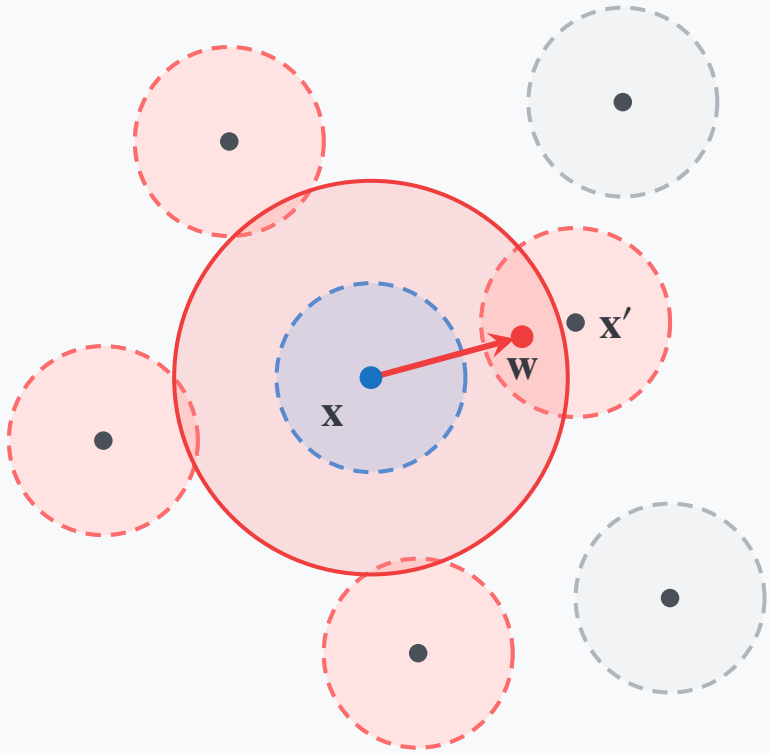
which contradicts the definition of $\mu(C)$. □



Unique Decoding Radius



(a) fewer than $\mu(C)/2$ errors



(b) more than $\mu(C)/2$ errors

- Balls of relative radius $\mu(C)/2$ around codewords never overlap. In (a), $\mathbf{w}$ stays in the ball of $\mathbf{x}$ and decodes to $\mathbf{x}$.
- In (b), $\mathbf{w}$ is closer to $\mathbf{x}'$: the decoder returns the wrong codeword.

$\mu(C)/2$ is called the **unique decoding radius** of C.

Example: Repetition Code

Example (Repetition code)

The repetition code over $\mathbb{F}_2$ of block length 3 repeats a bit three times: $C = \{000, 111\}$. Here $k = 1$ and $d_{\min}(C) = 3$.

- ✓ One bit flipped, $000 \mapsto 100$: still closer to 000 than to 111, since $d/2 = 3/2 > 1$.
- ✗ Two bits flipped, $000 \mapsto 110$: closer to 111, decoding fails, since two errors are not fewer than $d/2$.

Hamming Weight

Computing $d_{\min}(C)$ from the definition compares every pair. Practical codes offer a shortcut.

Definition (Hamming weight)

The **Hamming weight** $\|\mathbf{x}\|_0$ of $\mathbf{x} \in \Sigma^n$ is the number of non-zero symbols in $\mathbf{x}$.

When Σ is a field,

$$\Delta(\mathbf{u}, \mathbf{v}) = \|\mathbf{u} - \mathbf{v}\|_0,$$

that is, two vectors differ exactly in the coordinates where their difference is non-zero.

Linear Codes

Cryptographic codes are never arbitrary subsets of Σ^n: they have algebraic structure.

Definition (Linear code)

Let $\mathbb{F}_q$ be a finite field. A code $C \subseteq \mathbb{F}_q^n$ is a **linear code** if it is a linear subspace of $\mathbb{F}_q^n$. If C has dimension k and minimum distance d, we call it an $[n, k, d]_q$ code.

Question #3

Show that any linear encoding $\mathbb{F}^k \to C$ is $\mathbf{x} \mapsto \mathbf{G}\mathbf{x}$ for some $\mathbf{G} \in \mathbb{F}^{n \times k}$. What is rank($\mathbf{G}$)?

Distance Equals Weight

The difference of two codewords of a linear code is again a codeword, so the minimum over pairs collapses into a minimum over single codewords.

Proposition

Let $C \subseteq \mathbb{F}^n$ be a linear code. Then

$$\mu(C) = \frac{1}{n} \cdot \min_{0 \neq \mathbf{x} \in C} \|\mathbf{x}\|_0.$$

Proof idea.

$\leq$ $0 \in C$, and $\Delta(0, \mathbf{x}) = \|\mathbf{x}\|_0$.

$\geq$ If $\Delta(\mathbf{x}, \mathbf{y}) = d_{\min}(C)$, then $0 \neq \mathbf{x} - \mathbf{y} \in C$ and $\|\mathbf{x} - \mathbf{y}\|_0 = \Delta(\mathbf{x}, \mathbf{y})$. □

Rate

Definition (Rate)

Let $C \subseteq \mathbb{F}^n$ be a linear code of dimension k. The **rate** of C is $\rho(C) := k/n$.

- A codeword has n symbols but carries only k symbols of information: ρ is the fraction of actual payload.
- Rate and distance pull in opposite directions: every symbol is spent either on carrying the message or on keeping codewords apart.

Example: Parity Code

Example (Parity code)

The **parity code** consists of all vectors with an even number of ones:

$$C = \left\{ \mathbf{c} \in \mathbb{F}_2^n : c_1 + c_2 + \cdots + c_n = 0 \right\}.$$

To encode $\mathbf{m} \in \mathbb{F}_2^{n-1}$, append a **parity bit**. For $n = 4$: $101 \mapsto 1010$, $111 \mapsto 1111$.

- Linear, of dimension $n - 1$; weight 1 is impossible and $110\ldots0 \in C$, so $d_{\min}(C) = 2$.
- Hence C is an $[n, n-1, 2]_2$ code with rate $\rho(C) = 1 - 1/n$.
- A single error is **detected**, but not **corrected**: 1110 is at distance 1 from both 1010 and 1111.

Singleton Bound

To guarantee distance d, a code must dedicate at least $d-1$ of its n symbols to redundancy.

Proposition (Singleton bound)

Every $[n,k,d]_q$ code satisfies $k \leq n-d+1$, equivalently $\#C \leq q^{n-d+1}$.

Proof. Delete the first $d-1$ symbols of every codeword.

- Distinct codewords differ in $\geq d$ positions, so they still differ in one of the remaining $n-d+1$ positions.
- There are at most q^{n-d+1} such strings, so $q^k = \#C \leq q^{n-d+1}$. □

MDS Codes

Definition (MDS code)

An $[n, k, d]_q$ code is **maximum distance separable** (MDS) if $d = n - k + 1$, that is, if it attains the Singleton bound.

- No redundant symbol is wasted.
- Any k symbols of a codeword determine the whole codeword: the message survives any $n - k$ erasures.

Example

The repetition code $[n, 1, n]_2$ and the parity code $[n, n-1, 2]_2$ are MDS. A $[7, 4, 3]_2$ code is not MDS, since $n - k + 1 = 4 > 3$.

Question #4

Which of $[7, 4, 4]_q$, $[7, 4, 5]_q$, $[5, 1, 5]_q$, $[6, 3, 4]_q$ cannot be achieved by any linear code?

Reed–Solomon Codes

Idea: Encode into Polynomials

As with many ZK protocols, encode data into polynomials and store their **evaluations over a large domain**.

1: Put the message $(m_i)_{i\in[k]} \in \mathbb{F}^k$ into the coefficients: $f := \sum_{i\in[k]} m_i T^i \in \mathbb{F}[T]^{<k}$.
2: Fix an **evaluation domain** $\Omega = (\omega_i)_{i\in[n]}$ with $n > k$.
3: The codeword is $(f(\omega_i))_{i\in[n]}$.

Recall: Shamir's secret sharing

Decoding follows the same idea: if few evaluations are corrupted, we can still interpolate the polynomial f of degree $k - 1$.

Reed–Solomon Code

Restriction: for $f \in \mathbb{F}[T]^{<k}$ and a domain $\Omega \subseteq \mathbb{F}$, we write $\bar{f} : \Omega \to \mathbb{F}$ for f viewed only on inputs from Ω.

Remark

Feel free to treat $\bar{f}$ as the vector $(f(\omega_0), \ldots, f(\omega_{n-1}))$ for $\Omega := (\omega_0, \ldots, \omega_{n-1})$.

Definition (Reed–Solomon code)

The **Reed–Solomon code** over $\mathbb{F}$, evaluation domain $\Omega \subseteq \mathbb{F}$ (with $\#\Omega = n > k$), and degree k, is the linear code

$$\mathrm{RS}[\mathbb{F}, \Omega, k] := \{\bar{f} : \Omega \to \mathbb{F},\ f \in \mathbb{F}[T]^{<k}\}.$$

Example: $\mathrm{RS}[\mathbb{F}_7, \mathbb{F}_7, 3]$

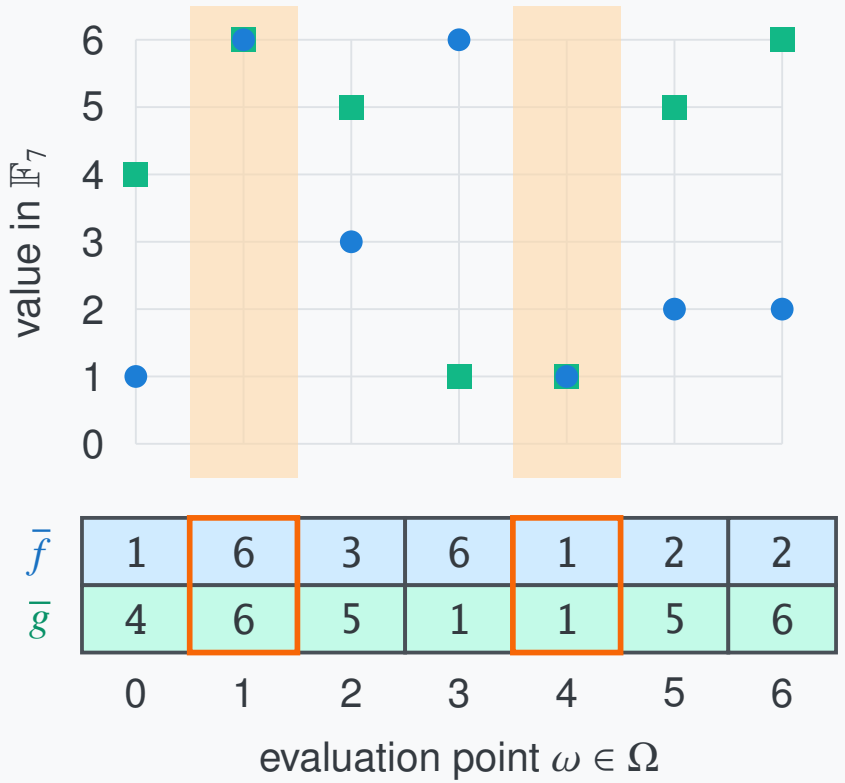
Take $\Omega = \mathbb{F}_7$, $n = 7$, $k = 3$.

- $(1, 2, 3) \mapsto f = 1 + 2T + 3T^2$,
- $(4, 0, 2) \mapsto g = 4 + 2T^2$.

Codewords:

$$\overline{f} = (1, 6, 3, 6, 1, 2, 2),$$
$$\overline{g} = (4, 6, 5, 1, 1, 5, 6).$$

They agree only at 1 and 4:
$f - g = (T - 1)(T - 4)$ has at most $k - 1 = 2$ roots, so $\overline{f}, \overline{g}$ differ in 5 of 7 positions.



Reed–Solomon Code is Linear

Proposition

RS$[\mathbb{F}, \Omega, k]$ is a linear code of length $n = \#\Omega$ and dimension k.

Proof.

- For $a \in \mathbb{F}$ and $f, g \in \mathbb{F}[T]^{<k}$, restriction commutes with both operations:
$$\overline{f} + \overline{g} = \overline{f + g}, \qquad a\overline{f} = \overline{af}.$$

- The map $f \mapsto \overline{f}$ is linear and injective: if $\overline{f} = 0$, then f has $n > k - 1$ roots, so $f = 0$. Hence the dimension is $\dim \mathbb{F}[T]^{<k} = k$. □

Reed–Solomon Codes are MDS

Proposition

RS$[\mathbb{F}, \Omega, k]$ with $n = \#\Omega$ and $q = \#\mathbb{F}$ is an $[n, k, n-k+1]_q$ code. That is, it attains the Singleton bound.

Proof. Fix distinct $f, g \in \mathbb{F}[T]^{<k}$. Then

$$\Delta(\overline{f}, \overline{g}) = n - \#\{\omega \in \Omega : f(\omega) = g(\omega)\} \geq n - (k-1),$$

since $f - g \neq 0$ has at most $k-1$ roots. The Singleton bound gives the other direction. □

Remark

The rate is $\rho = k/n$ and $\mu = 1 - \rho + 1/n$: lowering the rate directly increases the relative distance. Since n is large in practice, we use $\mu \approx 1 - \rho$.

Questions

Question #5

Consider RS$[\mathbb{F}_7, \Omega := \mathbb{F}_7, 2]$. What is the encoding of the message $(3, 2)$?

Question #6

Let $C \subseteq \mathbb{F}_2^4$ be the linear code spanned by $(1, 1, 0, 1)$ and $(0, 1, 1, 1)$. What is $d_{\min}(C)$?

Proximity to a Code

Distance to a Code

Classical coding theory: recover the original message. Our requirement is weaker:

- the verifier does not need to know **which** polynomial the prover committed to,
- only whether the committed vector is close to **some** codeword.

Definition (Distance to a code)

Let $C \subseteq \Sigma^n$ be a code and $\mathbf{w} \in \Sigma^n$. Then

$$\Delta(\mathbf{w}, C) := \min_{\mathbf{x} \in C} \Delta(\mathbf{w}, \mathbf{x}), \qquad \delta(\mathbf{w}, C) := \frac{1}{n} \Delta(\mathbf{w}, C).$$

$\mathbf{w}$ is δ-**close** to C if $\delta(\mathbf{w}, C) \leq \delta$, and δ-**far** otherwise.

The List of Close Codewords

Definition

Let C be an $[n, k, d]$ linear code. For $\mathbf{w} \in \mathbb{F}^n$ and $\delta \in [0, 1]$, define

$$\text{List}[\mathbf{w}, C, \delta] := \{\mathbf{x} \in C : \delta(\mathbf{x}, \mathbf{w}) \leq \delta\}.$$

In many cryptographic applications, we care about the **size** of this list.

Example

By unique decoding, if $\delta < \mu(C)/2$, then $\#\text{List}[\mathbf{w}, C, \delta] \leq 1$.

But...

For $\delta \geq \mu(C)/2$ the situation becomes very tricky, and our knowledge is quite limited.

Johnson Bound

Theorem (Johnson bound)

For a code $C \subseteq \mathbb{F}^n$, fixed $\mathbf{w} \in \mathbb{F}^n$, and $0 < \delta < 1 - \sqrt{1 - \mu(C)}$,

$$\#\mathrm{List}[\mathbf{w}, C, \delta] \leq 1/\varepsilon_\delta, \quad \varepsilon_\delta := 2\sqrt{1-\mu(C)}\,(1 - \sqrt{1-\mu(C)} - \delta).$$

- The bound only makes sense for $\delta < 1 - \sqrt{1-\mu(C)}$: $\varepsilon_\delta \to 0$ as δ approaches it.
- For $\delta \geq \mu(C)$, in the worst case the list is large.

Question #7

Let $C \subseteq \mathbb{F}^n$ be linear over a field of size q, and $\delta \geq \mu(C)$. Show that some $\mathbf{w} \in \mathbb{F}^n$ has $\#\mathrm{List}[\mathbf{w}, C, \delta] \geq q$.

Size of the List

What happens between $1 - \sqrt{1-\mu(C)}$ and $\mu(C)$? There is no general answer: it depends on the specifics of C.

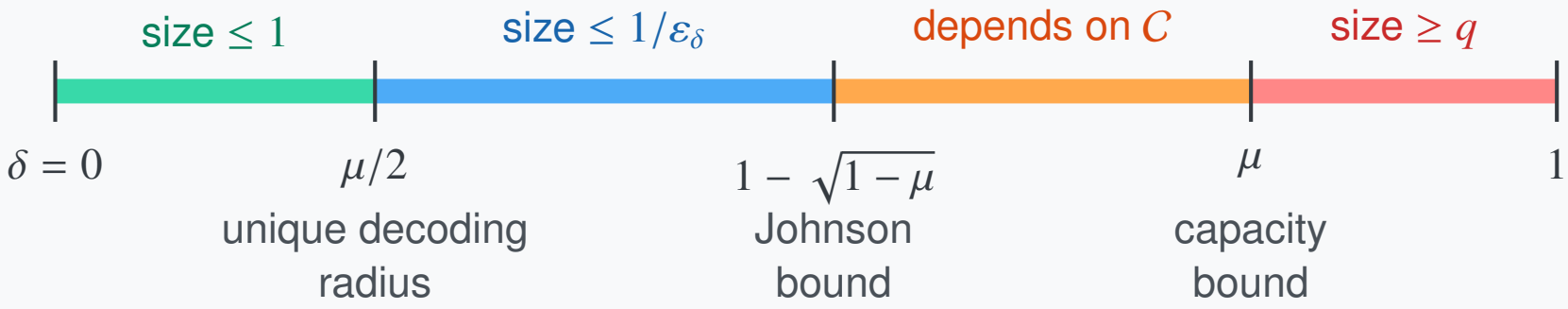


Figure: The size of List$[\mathbf{w}, C, \delta]$ as a function of $\delta \in [0,1]$, where $\mu = \mu(C)$. Adapted from Dan Boneh's ZK Whiteboard Session S2M7.

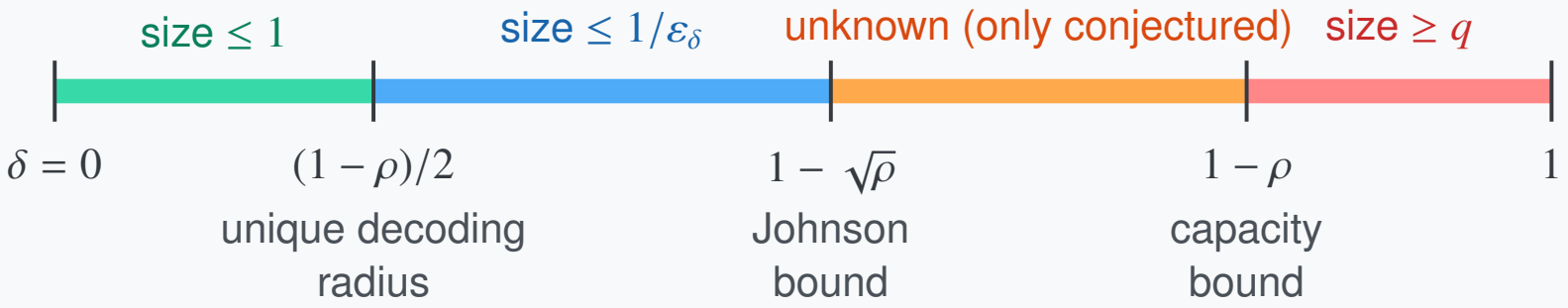
List Decoding for Reed–Solomon Codes

For $C = \mathrm{RS}[\mathbb{F}, \Omega, k]$, substitute $\mu(C) \approx 1 - \rho(C)$.

Proposition

Let $C = \mathrm{RS}[\mathbb{F}_q, \Omega, k]$, $\mathbf{w} \in \mathbb{F}^n$, and $\rho = \rho(C)$. Then

1. for $\delta \leq (1-\rho)/2$, $\#\mathrm{List}[\mathbf{w}, C, \delta] \leq 1$;
2. for $\delta < 1 - \sqrt{\rho}$, $\#\mathrm{List}[\mathbf{w}, C, \delta] \leq \varepsilon_\delta^{-1}$ for $\varepsilon_\delta := 2\sqrt{\rho}(1 - \sqrt{\rho} - \delta)$;
3. for $\delta \geq 1 - \rho$, $\#\mathrm{List}[\mathbf{w}, C, \delta] \geq q$.



The Proximity Prize

- The size of the list beyond the Johnson bound is so important that the Ethereum Foundation announced the **Proximity Prize** (proximityprize.org).
- Goal: verify or disprove the conjectures for larger δ's.
- A comprehensive review: survey by Gal Arnon, Dan Boneh, and Giacomo Fenzi.

Question #8

Consider RS$[\mathbb{F}_7, \Omega := \mathbb{F}_7, 2]$ and $\mathbf{w} := (0, 1, 4, 2, 2, 4, 1)$. What is $\delta(\mathbf{w}, \text{RS})$?

Hint: $\mathbf{w}$ is the evaluation of T^2 over $\Omega := \mathbb{F}_7$.

Why do we need List Decoding?

Assume that $\mathbf{w}$ is δ-far from C.

- Whichever $\mathbf{x} \in C$ we compare it to, they differ in more than δn positions.
- A uniformly random query detects a discrepancy with probability $> \delta$; t independent queries miss it with probability $< (1 - \delta)^t$.
- The larger the δ we can afford, the fewer queries and the shorter the proof.

The challenge

The verifier does not know in advance which codeword $\mathbf{x}$ it is checking against. An efficient test with few queries to $\mathbf{w}$ is exactly what the FRI protocol achieves.

Three Regimes

- **Unique decoding**, $\delta < (1-\rho)/2$: a query passes with probability $> (1+\rho)/2 > 1/2$, so less than one bit of security per query.
- Beyond this radius, $\mathbf{w}$ no longer determines a single codeword: every element of $\mathrm{List}[\mathbf{w}, C, \delta]$ is a candidate, and the soundness error grows with $\#\mathrm{List}[\mathbf{w}, C, \delta]$.
- **Johnson bound**, $\delta < 1 - \sqrt{\rho}$: the list is provably small, and a query passes with probability about $\sqrt{\rho}$.
- **Capacity**, $\delta < 1 - \rho$: the list is only **conjectured** to be small; protocols using this range rely on the conjecture.

Example

For $\rho = 1/4$ and 100 bits of security, the three regimes require roughly 148, 100, and 50 queries, respectively.

Beyond Proof Systems: Keys from Biometrics

Two readings $\mathbf{w}, \mathbf{w}' \in \mathbb{F}_2^n$ of the same face scan are never identical, only **close** in the Hamming distance, while a key must be reproduced exactly.

Code-offset fuzzy extractor

Let C be a binary $[n, k, 2t+1]_2$ code with a decoder correcting up to t errors, and H a hash function.

- Gen($\mathbf{w}$): sample $\mathbf{c} \leftarrow\!\!\$\ C$; output the public **helper string** $\mathbf{h} := \mathbf{c} + \mathbf{w}$ and the key $\mathsf{sk} := \mathsf{H}(\mathbf{c})$.
- Rep($\mathbf{h}, \mathbf{w}'$): decode $\mathbf{h} + \mathbf{w}'$ to a codeword $\mathbf{c}'$; output $\mathsf{sk}' := \mathsf{H}(\mathbf{c}')$.

Question #9

Show that $\mathsf{Rep}(\mathbf{h}, \mathbf{w}') = \mathsf{sk}$ whenever $\Delta(\mathbf{w}, \mathbf{w}') \leq t$. For a $[128, 44, 25]_2$ code, why does the key have at most 44 bits of security, however good the biometric is?

Summary

- A code $C \subseteq \Sigma^n$ adds redundancy; the **relative minimum distance** $\mu(C)$ governs error correction, and $\mu(C)/2$ is the unique decoding radius.
- **Linear** $[n, k, d]_q$ codes: distance equals minimum weight; rate $\rho = k/n$; Singleton bound $k \leq n - d + 1$.
- **Reed–Solomon** codes RS$[\mathbb{F}, \Omega, k]$: restrictions of polynomials of degree $< k$ to Ω; they are MDS and $\mu \approx 1 - \rho$.
- **Proximity**: the size of List$[\mathbf{w}, C, \delta]$ is ≤ 1 below $(1 - \rho)/2$, small up to the Johnson bound $1 - \sqrt{\rho}$, and only conjectured to be small up to $1 - \rho$.
- Next: testing proximity with few queries (IOPPs and FRI).

Thank you for your attention



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github.com/BlockstreamResearch/zk-book



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